Demystifying recalcitrant deep phylogenetic relationships within

Supplementary materials

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1 Assembling transcriptomes

```
## Assembling raw reads
```

```
Trinity --seqType fq --max_memory 50G --left $R1 --right $R2 --CPU 16 --output trinity_output

TransDecoder.LongOrfs -t $fasta
TransDecoder.Predict -t $fasta --single_best_only --cpu 4

## QC

busco -i $fasta -l ~/App/busco_db/metazoa_odb10/ -o busco_${fasta} -m prot -c 1 -f
```

2 Merging transcriptomes

```
species=`awk -F'\t' -v species_index="$i" '{if($1==species_index) print $2}'
filelist_species.tsv`

DATADIR=data/transcriptomic_assemblies/${species}

filelist=filelist_transcripto_all.tsv

assemblies=`awk -F"\t" -vspecies="$species" -vDATADIR="$DATADIR" -vOFS="/" '{if ($7==species)
print DATADIR,$5}' $filelist`  #| awk '{ORS=(NR%99?FS:RS)}1' `

echo "RUNNING SPECIES $i: $species"
echo "ASSEMBLIES= $assemblies"

## merge inputfiles

rm -rf ./scratch/trans_assembly/${species}/tr2aacds_3

mkdir -p ./scratch/trans_assembly/${species}/tr2aacds_3
cd ./scratch/trans_assembly/${species}/tr2aacds_3

cat $assemblies | sed 's/|/ /g' > tmp.transcripts.fasta

~/App/EvidentialGene/scripts/rnaseq/trformat.pl -output transcripts.fasta -input
tmp.transcripts.fasta

## run tr2aacds

~/App/EvidentialGene/scripts/prot/tr2aacds.pl -cdnaseq transcripts.fasta -logfile -runsteps
-noutrorf


## extract unique

cp okayset/transcripts.okay.aa ../${species}.merged.transcripts.okay.aa
cp okayset/transcripts.okay.cds ../${species}.merged.transcripts.okay.cds
cp okayset/transcripts.okay.mrna ../${species}.merged.transcripts.okay.mrna


~/App/seqkit grep -r -n -p 't1 ' okayset/transcripts.okay.aa >
../${species}.merged.transcripts.okay.unique.aa

## run busco
```

```
conda activate ~/.conda/envs/mamba/envs/busco_v550

busco -i ../${species}.merged.transcripts.okay.aa -l ~/App/busco_db/metazoa_odb10/ -o
busco_transcripts.okay.aa -m prot -c 1 -f
cp busco_transcripts.okay.aa/run_metazoa_odb10/short_summary.txt
../${species}.merged.transcripts.okay.aa.busco

busco -i ../${species}.merged.transcripts.okay.unique.aa -l ~/App/busco_db/metazoa_odb10/ -o
busco_transcripts.okay.unique.aa -m prot -c 1 -f
cp busco_transcripts.okay.unique.aa/run_metazoa_odb10/short_summary.txt
../${species}.merged.transcripts.okay.unique.aa.busco

id=80
cd-hit -i ../${species}.merged.transcripts.okay.unique.aa -o
../${species}.merged.transcripts.okay.unique.cdhit${id}.pep -c 0.${id} -M 4000 -T 1

busco -i ../${species}.merged.transcripts.okay.unique.cdhit${id}.pep -l
~/App/busco_db/metazoa_odb10/ -o busco_transcripts.okay.unique.cdhit${id}.pep -m prot -c 1 -f
cp busco_transcripts.okay.unique.cdhit${id}.pep/run_metazoa_odb10/short_summary.txt
../${species}.merged.transcripts.okay.unique.cdhit${id}.pep.busco


## get summary of buscos

if [[ -f ../${species}.merged.transcripts.okay.unique.aa ]] && [[ -f
../${species}.merged.transcripts.okay.unique.aa ]]; then

echo -e "SUCCESS\t${i}\t${species}\t$(date '+%Y-%m-%d')\t$(grep 'C:'
../${species}.merged.transcripts.okay.unique.cdhit${id}.pep.busco | sed 's/\t//g')"  >>
work/Trans_assembly/summary_tr2aacds_processed.txt

else

echo -e "FAIL\t${i}\t${species}\t$(date '+%Y-%m-%d')\tNA" >>
work/Trans_assembly/summary_tr2aacds_processed.txt

fi
```

3 Finding gene families (orthogroups)

```
# Finding orthogroups with Orthofinder

datadir=144species #A31 for reduced dataset

orthofinder -f $datadir -t $SLURM_CPUS_PER_TASK -a $SLURM_CPUS_PER_TASK -S diamond -T fasttree
-ot -I 1.5 -M msa -z
```

4 Splitting gene families into single-copy orthologs

```
CPUS=$1
```

```
source ~/.bashrc
mamba activate orthosnap
export PERL5LIB= PERL_LOCAL_LIB_ROOT= cpan

mkdir -p ./OrthoFinder/Orthosnap
cd ./OrthoFinder/Orthosnap

orthosnap () {

OG=$1

mkdir -p snaps
mkdir -p out

orthosnap -t input_trees/${OG}.tree -f input_aln/${OG}.fa -op snaps/ -o 4 > out/${OG}.out


##### summary of the results

if [ -s snaps/${OG}.fa.orthosnap.0.fa ]; then

echo -e "$OG\tsnapped" >> OG_orthosnap_result.txt

else
if [ "$(tail -n2 out/$OG.out | head -n1)" == "Input phylogeny is already a single-copy
orthogroup" ]; then

echo -e "$OG\tnot" >> OG_orthosnap_result.txt
ln -s aln/${OG}.fa snaps/${OG}.fa # make a soft copy to the snaps folder

else
echo -e "$OG\terror" >> OG_orthosnap_result.txt
fi
fi

}

export -f orthosnap


parallel -j $CPUS orthosnap {1} :::: OG.filelist
```

5 Computing gene trees

5.1 Realigning and trimming

```
mkdir -p trimmed
mkdir -p trimal_out
mkdir -p phykit

trim () {
```

```
file=$1

### trim
## remove columns with > 90% gaps
## residue must be present in >50% of sequences and at least 30% of residues must pass the
threshold

trimal -in ../Orthosnap/input_aln/${file}.fa -out trimmed/${file}.faa -keepheader -fasta -gt
0.1 -resoverlap 0.5 -seqoverlap 30 > trimal_out/${file}.out

if [[ ! -s "trimal_out/${file}.out" ]]; then
rm trimal_out/${file}.out
fi

if [[ -s "trimmed/${file}.faa" ]]; then

echo $file | awk -vOFS="\t" '{print $1,"ok"}' >> trimmal_results.txt

## phykit1
tax_occ=`grep -E "^>[0-9]+|[0-9]+$" trimmed/${file}.faa | awk -F"|" '{print $1}' | sort | uniq
| wc -l`
seqcount=`grep ">" trimmed/${file}.faa | wc -l`

echo -e "$file\t$tax_occ\t$seqcount" >> phykit/tax_occ.tsv

pk_rcv trimmed/${file}.faa | awk -F'\t' -vOFS='\t' -valn="$file" '{print aln,$0}' >>
phykit/pk_rcv.tsv

pk_pis trimmed/${file}.faa | awk -F'\t' -vOFS='\t' -valn="$file" '{print aln,$0}' >>
phykit/pk_pis.tsv

pk_aln_len_no_gaps trimmed/${file}.faa | awk -F'\t' -vOFS='\t' -valn="$file" '{print
aln,$1,$2,$3}' >> phykit/pk_aln_len_no_gaps.tsv


else
echo $file | awk -vOFS="\t" '{print $1,"error"}' >> trimmal_results.txt
fi


}

export -f trim

parallel --jobs $CPUS trim {1} :::: OG_genes.list
parallel --jobs $CPUS trim {1} :::: SOG_genes.list
```

5.2 Computing OG trees (LG/WAG + tree)

```
run_fastree_LG () {
SUPERMATRIX=$1
ALIGNMENT=$2
OG=`echo $ALIGNMENT | awk -F'[/|.]' '{print $2}'`
```

```
TREE=${SUPERMATRIX}_trees/${OG}.tree

FastTree -lg -gamma -pseudo $ALIGNMENT > $TREE


if [ -s "$TREE" ] && [ -s "$ALIGNMENT" ]; then
echo '' | awk -vOFS="\t" -vOG="$OG" '{print "success",OG}' >> remaining_alignments.txt
else
echo '' | awk -vOFS="\t" -vOG="$OG" '{print "fail",OG}' >> remaining_alignments.txt
fi

}


run_fastree_wag () {
SUPERMATRIX=$1
ALIGNMENT=$2
OG=`echo $ALIGNMENT | awk -F'[/|.]' '{print $2}'`
TREE=${SUPERMATRIX}_trees/${OG}.tree

FastTree -wag -gamma -pseudo $ALIGNMENT > $TREE


if [ -s "$TREE" ] && [ -s "$ALIGNMENT" ]; then
echo '' | awk -vOFS="\t" -vOG="$OG" '{print "success",OG}' >> remaining_alignments.txt
else
echo '' | awk -vOFS="\t" -vOG="$OG" '{print "fail",OG}' >> remaining_alignments.txt
fi

}


export -f run_fastree_lg
export -f run_fastree_wag

parallel --jobs $CPUS run_fastree_lg "$SUPERMATRIX" {} :::: OG_genes.list


parallel --jobs $CPUS run_fastree_wag "$SUPERMATRIX" {} :::: large_OG_genes.list
```

5.3 Computing SOG trees (site homogeneous model selection + tree)

```
pargenes.py -c $CPUS --alignments-dir ${SUPERMATRIX}_data --output-dir ${SUPERMATRIX}_working
-d aa -m --bs-trees 100 --parsimony-starting-trees 5 --random-starting-trees 5 --continue


# Extract trees and gene models

python ~/.conda/envs/mamba/envs/pargenes/bin/pargenes_src/export.py -o ${SUPERMATRIX}_results
-i ${SUPERMATRIX}_working --support-values-tree --best-ml-model

cd ${SUPERMATRIX}_results
```

```
for file in `find . -name '*.raxml.bestModel'`
do

OG=`echo $file | awk -F'[./]' '{print $3}' | sed 's/_/./g'`
sed "s/noname/$OG/g" $file >> ../${SUPERMATRIX}.bestModels

TREE=$(echo $file | sed 's/.raxml.bestModel/.support.raxml.support/g')
cat $TREE >> ../${SUPERMATRIX}.genetrees.nx

echo "${OG}" >> ../${SUPERMATRIX}.genetrees.nx.ids

done
```

```
mkdir -p C60_trees

########################################

# -mem option does not work with partition models yet

run_C60 () {

OG_aln=$1
OG2=$(echo "${OG_aln}" | awk -F'.' '{print $1}')

tree_file=scratch/tree_building/OG75/LG_coalescent/OG75_trees/${OG2}.tree
aln_file=scratch/tree_building/OG75/OG75_data/${OG_aln}
output_file=C60_trees/${OG2}.treefile

#echo $tree_file, $aln_file, $output_file

# Check if the output file already exists
if [ -f "$output_file" ]; then
    echo "$OG2 already processed, skipping."
    echo "$OG2" >> finished_processing.txt
    exit 0
fi

# Run iqtree2 if the output file does not exist
iqtree2 -s "$aln_file" -m LG+C60+F+G -ft "$tree_file" --seqtype AA --prefix C60_trees/${OG2}
-ntmax 1 -nt 1

# Check if iqtree2 ran successfully and the output file was created
if [ -f "$output_file" ]; then
    echo "$OG2" >> finished_processing.txt
else
    echo "$OG2" >> error_processing.txt
fi

}

export -f run_C60

ls -lS ./scratch/tree_building/${SUPERMATRIX}/OG75_data/ | awk '{print $NF}' | tail -n+2 | tac
> OG75.size_sorted.filelist
```

```
parallel --jobs $CPUS run_C60 {} :::: OG_genes.list
parallel --jobs $CPUS run_C60 {} :::: SOG_genes.list
```

5.3.0.1 Computing trees under site heterogeneous model

5.4 Computing gene trees under site heterogeneous model

```
mkdir -p C60_trees

run_C60 () {

OG_aln=$1
OG2=$(echo "${OG_aln}" | awk -F'.' '{print $1}')

tree_file=${OG2}.tree
aln_file=${OG_aln}
output_file=C60_trees/${OG2}.treefile

# Check if the output file already exists
if [ -f "$output_file" ]; then
    echo "$OG2 already processed, skipping."
    echo "$OG2" >> finished_processing.txt
    exit 0
fi

# Run iqtree2 if the output file does not exist
iqtree2 -s "$aln_file" -m LG+C60+F+G -ft "$tree_file" --seqtype AA --prefix C60_trees/${OG2}
-ntmax 1 -nt 1

# Check if iqtree2 ran successfully and the output file was created
if [ -f "$output_file" ]; then
    echo "$OG2" >> finished_processing.txt
else
    echo "$OG2" >> error_processing.txt
fi

}

export -f run_C60

parallel --jobs $CPUS run_C60 {} :::: gene_trees.filelist
```

5.5 Tree statistics and submatrices

```
phykit () {

SUPERMATRIX=$1
file=$2
file2=`echo $file | awk -F'[/|.]' '{print $2".tree"}'`

echo "$SUPERMATRIX $file $file2"
```

```
tax_occ=`grep -E "^>[0-9]+|[0-9]+$" ${SUPERMATRIX}_data/$file | awk -F"|" '{print $1}' | sort |
uniq | wc -l`
seqcount=`grep ">" ${SUPERMATRIX}_data/$file | wc -l`
echo -e "$file\t$tax_occ\t$seqcount" >> phykit/tax_occ.tsv

pk_rcv ${SUPERMATRIX}_data/$file | awk -F'\t' -vOFS='\t' -valn="$file" '{print aln,$0}' >>
phykit/pk_rcv.tsv

pk_pis ${SUPERMATRIX}_data/$file | awk -F'\t' -vOFS='\t' -valn="$file" '{print aln,$0}' >>
phykit/pk_pis.tsv

pk_evo_rate ${SUPERMATRIX}_trees/${file2} | awk -F'\t' -vOFS='\t' -valn="$file" '{print
aln,$0}' >> phykit/pk_evo_rate.tsv

pk_aln_len_no_gaps ${SUPERMATRIX}_data/$file | awk -F'\t' -vOFS='\t' -valn="$file" '{print
aln,$1,$2,$3}' >> phykit/pk_aln_len_no_gaps.tsv

pk_lb_score ${SUPERMATRIX}_trees/${file2} | awk -F': ' -vOFS='\t' -valn="$file"
'{if($1=="standard deviation") print aln,$2}' >> phykit/pk_dslb_score.tsv

pk_lb_score ${SUPERMATRIX}_trees/${file2} | awk -F': ' -vOFS='\t' -valn="$file"
'{if($1=="mean") print aln,$2}' >> phykit/pk_alb_score.tsv

pk_saturation -a ${SUPERMATRIX}_data/$file -t ${SUPERMATRIX}_trees/${file2} | awk -F'\t'
-vOFS='\t' -valn="$file" '{print aln,$0}' >> phykit/pk_saturation.tsv

pk_tness ${SUPERMATRIX}_trees/${file2} | awk -F'\t' -vOFS='\t' -valn="$file" '{print aln,$0}'
>> phykit/pk_tness.tsv

pk_tor -a ${SUPERMATRIX}_data/$file -t ${SUPERMATRIX}_trees/${file2} | awk -F'\t' -vOFS='\t'
-valn="$file" '{print aln,$0}' >> phykit/pk_tor.tsv

pk_bss ${SUPERMATRIX}_trees/${file2} | awk -F': ' -vOFS='\t' -valn="$file" '{if($1=="mean")
print aln,$2}' >> phykit/pk_abs.tsv

}

export -f phykit

mkdir -p phykit


~/.conda/envs/mamba/bin/parallel --jobs $CPUS phykit $SUPERMATRIX {1} ::::
${SUPERMATRIX}.filelist
```

6 Computing species trees

6.1 Contatenation approach

6.1.1 Edge proportional partitionned model

```
iqtree2 -s ${SUPERMATRIX}_data -p data/gene_alignments/${SUPERMATRIX}.partition.nex --seqtype
AA --prefix EDGEPROP.${SUPERMATRIX} -ntmax $CPUS -nt $CPUS -B 1000  --alrt 1000
```

6.1.2 Site heterogeneous model

```
CPUS=$1
SUPERMATRIX=$2
TOPO=SITEHET.${SUPERMATRIX}

TMPDIR=./scratch/tree_building/${SUPERMATRIX}
mkdir -p $TMPDIR
cd $TMPDIR

mamba activate catfasta2phyml
AMAS.py concat -p ${SUPERMATRIX}.partitions.nex.txt -t ${SUPERMATRIX}.concatenated.fa -u fasta
-y nexus -i ${SUPERMATRIX}_data/* -f fasta -d aa
mamba deactivate

iqtree2 -s ${SUPERMATRIX}.concatenated.fa -m LG+C60+F+G -ft EDGEPROP.${SUPERMATRIX}.treefile
--seqtype AA --prefix ${TOPO} -ntmax $CPUS -nt $CPUS -mem 600G -n 0 -redo

iqtree2 -s ${SUPERMATRIX}.concatenated.fa -m LG+C60+F+G -fs ${TOPO}.sitefreq --seqtype AA
--prefix ${TOPO} -ntmax $CPUS -nt $CPUS -B 1000 -alrt 1000 --undo
```

6.2 Reconciliation approach

```
mpiexec -np $CPUS asteroid --input-gene-trees ${SUPERMATRIX}.genetrees.nx
--gene-species-mapping ${SUPERMATRIX}.genetrees.map.txt --min-bl 0.01 --bs-replicates 100
--prefix ${SUPERMATRIX}.asteroid
```

6.3 Computing constrained trees

```
#!/bin/r, echo = TRUE, eval=FALSE

CPUS=$1
TOPO=$2

TMPDIR=./scratch/tree_building/constrained
mkdir -p $TMPDIR
cd $TMPDIR

### Site homogeneous ###

iqtree2 -S ${SUPERMATRIX}_data -p ./data/gene_alignments/${SUPERMATRIX}.partition.nex --seqtype
AA --prefix EDGEPROP.${SUPERMATRIX}.constrained_to_${TOPO} -ntmax $CPUS -nt $CPUS -te
./data/trees/mycladetopo.${TOPO}.tree

### Site heterogeneous ###

iqtree2 -s ${SUPERMATRIX}.concatenated.fa -m LG+C60+F+G -ft EDGEPROP.${SUPERMATRIX}.treefile
--seqtype AA --prefix SITEHET.${SUPERMATRIX}.constrained_to_${TOPO} -ntmax $CPUS -nt $CPUS -mem
600G -n 0 -te ./data/trees/mycladetopo.${TOPO}.tree
```

```
iqtree2 -s ${SUPERMATRIX}.concatenated.fa -m LG+C60+F+G -ft EDGEPROP.${SUPERMATRIX}.treefile
--seqtype AA --prefix SITEHET.${SUPERMATRIX}.constrained_to_${TOPO} -ntmax $CPUS -nt $CPUS -B
1000 -alrt 1000 --scfl 100 --undo

####################
treels () {
cat ./scratch/tree_building/${SUPERMATRIX}/EDGEPROP.${SUPERMATRIX}.treefile >
EDGEPROP.${SUPERMATRIX}.constrained_to_topo_d.treels

for TOPO in topoM1d topoM2d topoM3d topoM4d topoM5d topoC1d topoM1d topoC3d topoC4d topoS1d
topoS2d topoM1d topoS4d topoS5d

do
cat EDGEPROP.${SUPERMATRIX}.constrained_to_${TOPO}.treefile >>
EDGEPROP.${SUPERMATRIX}.constrained_to_topo_d.treels

done

cp EDGEPROP.${SUPERMATRIX}.constrained_to_topo_d.treels
$MYDIR/EDGEPROP.${SUPERMATRIX}.constrained_to_topo_d.treels
}

#########
post_processing () {
grep -B 5 "BIC" ./scratch/tree_building/${SUPERMATRIX}/EDGEPROP.${SUPERMATRIX}.iqtree | awk
-F'\t' -vOFS='\t' '{print "SOG10",$0}' >
EDGEPROP.${SUPERMATRIX}.constrained_to_topo_d.tree_scores


for TOPO in topoM1d topoM2d topoM3d topoM4d topoM5d topoC1d topoM1d topoC3d topoC4d topoS1d
topoS2d topoM1d topoS4d topoS5d
do

grep -B 5 "BIC" EDGEPROP.${SUPERMATRIX}.constrained_to_${TOPO}.iqtree | awk -F'\t' -vOFS='\t'
-vtopo="$TOPO" '{print topo,$0}' >> EDGEPROP.${SUPERMATRIX}.constrained_to_topo_d.tree_scores

done

cp EDGEPROP.${SUPERMATRIX}.constrained_to_topo_d.tree_scores $MYDIR/.
}
```

7 Assessing support

7.1 For species trees

```
CPUS=$1

SUPERMATRIX=SOG10
MODEL=EDGEPROP

GENE_TREES=./scratch/tree_building/${SUPERMATRIX}/${SUPERMATRIX}.genetrees.renamed.nx
```

```
cd ./scratch/tree_building/${SUPERMATRIX}

#get qCF for SOG
astral4 --root "87" -C -t $CPUS -u 3 -q ${MODEL}.${SUPERMATRIX}.treefile -i ${GENE_TREES} -t 2
-o ${MODEL}.${SUPERMATRIX}.astral_species_annotated.tree 2>
${MODEL}.${SUPERMATRIX}.astral_species_annotated.log

#Get sCF
iqtree2 -te ${MODEL}.${SUPERMATRIX}.astral_species_annotated.tree --scfl 100 -s
./scratch/tree_building/${SUPERMATRIX}/${SUPERMATRIX}_data -p
./scratch/tree_building/${SUPERMATRIX}/${SUPERMATRIX}.partition.nex  --prefix
${MODEL}.${SUPERMATRIX}.scfl -T $CPUS

#Get gCF
iqtree2 -te ${MODEL}.${SUPERMATRIX}.astral_species_annotated.tree --gcf ${GENE_TREES} --prefix
${MODEL}.${SUPERMATRIX}.gcf -T $CPUS

#Get branch lengths in coalescent units
iqtree2 -te ${MODEL}.${SUPERMATRIX}.astral_species_annotated.tree -blfix -p
./scratch/tree_building/${SUPERMATRIX}/${SUPERMATRIX}.partition.nex --scfl 1 --prefix
${MODEL}.${SUPERMATRIX}.coalescent_bl -T $CPUS
```

```
CPUS=$1

SUPERMATRIX=OG75
MODEL=MAT

GENE_TREES=./scratch/tree_building/${SUPERMATRIX}/${SUPERMATRIX}.genetrees.renamed.nx

cd ./scratch/tree_building/${SUPERMATRIX}

#get qCF for OG

GENE_TREES=./data/trees/.${SUPERMATRIX}.genetrees.nx
GENE_MAP=./data/trees/.${SUPERMATRIX}.genetrees.seqlist.txt

#root "22" for A31 dataset
astral-pro --root "87" -R -t $CPUS -u 3 --genelength 500 -o
${MODEL}.${SUPERMATRIX}.astral-pro3_species_annotated.tree -i ${GENE_TREES} -a ${GENE_MAP} 2>
${MODEL}.${SUPERMATRIX}.astral-pro3_species_annotated.log
```

7.2 For constrained trees

```
CPUS=$1
TOPO=$2

TMPDIR=./scratch/tree_building/constrained
mkdir -p $TMPDIR
cd $TMPDIR



run_C60 () {
```

```
## SOG10_all

SOG10_GENE_TREES=./data/trees/SOG10/SOG10.genetrees.renamed.nx

# MAT qCF
astral4 --root "87" -C -t $CPUS -u 3 -c ./data/trees/mycladetopo.${TOPO}.tree -o
EDGEPROP.SOG10_all.constrained_to_${TOPO}.astral4_species_annotated.tree -i ${SOG10_GENE_TREES}
2> EDGEPROP.SOG10_all.constrained_to_${TOPO}.astral4_species_annotated.log

mv freqQuad.csv
EDGEPROP.SOG10_all.constrained_to_${TOPO}.astral4_species_annotated.freqQuad.csv

#MAT gCF
iqtree2 -te EDGEPROP.SOG10_all.constrained_to_${TOPO}.astral4_species_annotated.tree --gcf
${SOG10_GENE_TREES} --prefix EDGEPROP.SOG10_all.constrained_to_${TOPO}.gcf.astral4 -T $CPUS

### SOG10_C60 ###
#C60 qCF
astral4 --root "87" -C -t $CPUS -u 3 -c ./data/trees/mycladetopo.${TOPO}.tree -a
../SOG10_C60/SOG10_C60.genetrees.map.txt -o
EDGEPROP.SOG10_C60.constrained_to_${TOPO}.astral4_species_annotated.tree -i
../SOG10_C60/SOG10_C60.genetrees.nx 2>
EDGEPROP.SOG10_C60.constrained_to_${TOPO}.astral4_species_annotated.log

mv freqQuad.csv
EDGEPROP.SOG10_C60.constrained_to_${TOPO}.astral4_species_annotated.freqQuad.csv

#C60 gCF
iqtree2 -te EDGEPROP.SOG10_C60.constrained_to_${TOPO}.astral4_species_annotated.tree --gcf
${SOG10_GENE_TREES} --prefix EDGEPROP.SOG10_C60.constrained_to_${TOPO}.gcf.astral4 -T $CPUS
```

7.3 Topology tests for constrained trees

7.3.1 Site homogeneous

```
CPUS=$1
TMPDIR=./scratch/tree_building/constrained
mkdir -p $TMPDIR
cd $TMPDIR

SUPERMATRIX=SOG10_all

#early branching Annelida
make_treels(){
rm EDGEPROP.${SUPERMATRIX}.constrained_to_topo_Ad.treels
for TOPO in topoA1d topoA2d topoA3d
do
cat EDGEPROP.${SUPERMATRIX}.constrained_to_${TOPO}.treefile >>
EDGEPROP.${SUPERMATRIX}.constrained_to_topo_Ad.treels
done

iqtree2 -S ${SUPERMATRIX}_data -p ./data/gene_alignments/${SUPERMATRIX}.partition.nex --seqtype
AA --prefix EDGEPROP.${SUPERMATRIX}.constrained_to_topo_Ad -te
EDGEPROP.${SUPERMATRIX}.constrained_to_topoM1d.treefile -ntmax $CPUS -nt $CPUS -z
EDGEPROP.${SUPERMATRIX}.constrained_to_topo_Ad.treels -n 0 -zb 10000 -zw -au -redo
```

```
}

#export -f make_treels
#make_treels

# Myzo
make_treels(){
# REF=M1d

rm EDGEPROP.${SUPERMATRIX}.constrained_to_topo_Md.treels
for TOPO in topoM2d topoM3d topoM4d topoM5d
do
cat EDGEPROP.${SUPERMATRIX}.constrained_to_${TOPO}.treefile >>
EDGEPROP.${SUPERMATRIX}.constrained_to_topo_Md.treels
done

iqtree2 -S ${SUPERMATRIX}_data -p ./data/gene_alignments/${SUPERMATRIX}.partition.nex --seqtype
AA --prefix EDGEPROP.${SUPERMATRIX}.constrained_to_topo_Md -te
EDGEPROP.${SUPERMATRIX}.constrained_to_topoM1d.treefile -ntmax $CPUS -nt $CPUS -z
EDGEPROP.${SUPERMATRIX}.constrained_to_topo_Md.treels -n 0 -zb 10000 -zw -au -redo

}

export -f make_treels
make_treels


#Clit
make_treels(){
rm EDGEPROP.${SUPERMATRIX}.constrained_to_topo_Cd.treels
for TOPO in topoC1d topoC3d topoC4d
do
cat EDGEPROP.${SUPERMATRIX}.constrained_to_${TOPO}.treefile >>
EDGEPROP.${SUPERMATRIX}.constrained_to_topo_Cd.treels
done

iqtree2 -S ${SUPERMATRIX}_data -p ./data/gene_alignments/${SUPERMATRIX}.partition.nex --seqtype
AA --prefix EDGEPROP.${SUPERMATRIX}.constrained_to_topo_Cd -te
EDGEPROP.${SUPERMATRIX}.constrained_to_topoM1d.treefile -ntmax $CPUS -nt $CPUS -z
EDGEPROP.${SUPERMATRIX}.constrained_to_topo_Cd.treels -n 0 -zb 10000 -zw -au -redo

}

export -f make_treels
make_treels

#SibCir
make_treels(){
rm EDGEPROP.${SUPERMATRIX}.constrained_to_topo_Sd.treels
for TOPO in topoS1d topoS2d topoS4d topoS5d
do
cat EDGEPROP.${SUPERMATRIX}.constrained_to_${TOPO}.treefile >>
EDGEPROP.${SUPERMATRIX}.constrained_to_topo_Sd.treels
done
```

```
iqtree2 -S ${SUPERMATRIX}_data -p ./data/gene_alignments/${SUPERMATRIX}.partition.nex --seqtype
AA --prefix EDGEPROP.${SUPERMATRIX}.constrained_to_topo_Sd -te
EDGEPROP.${SUPERMATRIX}.constrained_to_topoM1d.treefile -ntmax $CPUS -nt $CPUS -z
EDGEPROP.${SUPERMATRIX}.constrained_to_topo_Sd.treels -n 0 -zb 10000 -zw -au -redo

}

export -f make_treels
make_treels


############################################################
# All: comparing constrained trees to unconstrained tree
############################################################

rm EDGEPROP.${SUPERMATRIX}_to_topo_d.treels

for TOPO in topoA1d topoA2d topoA3d topoM1d topoM2d topoM3d topoM4d topoM5d topoC1d topoC3d
topoC4d topoS1d topoS2d topoS4d topoS5d
do
cat EDGEPROP.${SUPERMATRIX}.constrained_to_${TOPO}.treefile >>
EDGEPROP.${SUPERMATRIX}_to_topo_d.treels

done

iqtree2 -S ${SUPERMATRIX}_data -p ./data/gene_alignments/${SUPERMATRIX}.partition.nex --seqtype
AA --prefix EDGEPROP.${SUPERMATRIX}.constrained_to_topo_d -te
./scratch/tree_building/${SUPERMATRIX}/EDGEPROP.${SUPERMATRIX}.treefile -ntmax $CPUS -nt $CPUS
-z EDGEPROP.${SUPERMATRIX}.constrained_to_topo_d.treels -n 0 -zb 10000 -zw -au -redo
```

7.3.2 Site heterogeneous

```
#!/bin/bash

source ~/.bashrc

SUPERMATRIX=SOG10_tax_occ

CPUS=$1
TMPDIR=./scratch/tree_building/${SUPERMATRIX}
mkdir -p $TMPDIR
cd $TMPDIR


REF_ALN=./scratch/tree_building/${SUPERMATRIX}/${SUPERMATRIX}.concatenated.fa
REF_SITEFREQ=./scratch/tree_building/${SUPERMATRIX}/SITEHET.${SUPERMATRIX}.sitefreq

# Myzo
make_treels(){

rm SITEHET.${SUPERMATRIX}.constrained_to_topo_Md.treels
for TOPO in topoM2d topoM3d topoM4d topoM5d
do
cat SITEHET.${SUPERMATRIX}.constrained_to_${TOPO}.treefile >>
SITEHET.${SUPERMATRIX}.constrained_to_topo_Md.treels
```

```
done


iqtree2 -s $REF_ALN -m LG+C60+F+G -fs $REF_SITEFREQ --seqtype AA --prefix
SITEHET.${SUPERMATRIX}.constrained_to_topo_Md -te
SITEHET.${SUPERMATRIX}.constrained_to_topoM1d.treefile -ntmax $CPUS -nt $CPUS -z
SITEHET.${SUPERMATRIX}.constrained_to_topo_Md.treels -n 0 -zb 10000 -zw -au --undo

iqtree2 -s $REF_ALN -m LG+C60+F+G -fs $REF_SITEFREQ --seqtype AA --prefix
SITEHET.${SUPERMATRIX}.constrained_to_topo_Md -te
SITEHET.${SUPERMATRIX}.constrained_to_topoM1d.treefile -ntmax $CPUS -nt $CPUS -z
SITEHET.${SUPERMATRIX}.constrained_to_topo_Md.treels -n 0 -zb 10000 -zw -au  --undo
}

export -f make_treels
make_treels


#Clit
make_treels(){
rm SITEHET.${SUPERMATRIX}.constrained_to_topo_Cd.treels
for TOPO in topoC1d topoC3d topoC4d
do
cat SITEHET.${SUPERMATRIX}.constrained_to_${TOPO}.treefile >>
SITEHET.${SUPERMATRIX}.constrained_to_topo_Cd.treels
done

iqtree2 -s $REF_ALN -m LG+C60+F+G -fs $REF_SITEFREQ --seqtype AA --prefix
SITEHET.${SUPERMATRIX}.constrained_to_topo_Cd -te
SITEHET.${SUPERMATRIX}.constrained_to_topoM1d.treefile -ntmax $CPUS -nt $CPUS -z
SITEHET.${SUPERMATRIX}.constrained_to_topo_Cd.treels -n 0 -zb 10000 -zw -au --undo

}

export -f make_treels
make_treels

#SibCir
make_treels(){
rm SITEHET.${SUPERMATRIX}.constrained_to_topo_Sd.treels
for TOPO in topoS1d topoS2d topoS4d topoS5d
do
cat SITEHET.${SUPERMATRIX}.constrained_to_${TOPO}.treefile >>
SITEHET.${SUPERMATRIX}.constrained_to_topo_Sd.treels
done


iqtree2 -s $REF_ALN -m LG+C60+F+G -fs $REF_SITEFREQ --seqtype AA --prefix
SITEHET.${SUPERMATRIX}.constrained_to_topo_Sd -te
SITEHET.${SUPERMATRIX}.constrained_to_topoM1d.treefile -ntmax $CPUS -nt $CPUS -z
SITEHET.${SUPERMATRIX}.constrained_to_topo_Sd.treels -n 0 -zb 10000 -zw -au --undo

}

export -f make_treels
make_treels
```

```
# Early-Branching Annelida

make_treels(){
rm SITEHET.${SUPERMATRIX}.constrained_to_topo_Ad.treels
for TOPO in topoA1d topoA2d topoA3d
do
cat SITEHET.${SUPERMATRIX}.constrained_to_${TOPO}.treefile >>
SITEHET.${SUPERMATRIX}.constrained_to_topo_Ad.treels
done


iqtree2 -s $REF_ALN -m LG+C60+F+G -fs $REF_SITEFREQ --seqtype AA --prefix
SITEHET.${SUPERMATRIX}.constrained_to_topo_Ad -te
SITEHET.${SUPERMATRIX}.constrained_to_topoM1d.treefile -ntmax $CPUS -nt $CPUS -z
SITEHET.${SUPERMATRIX}.constrained_to_topo_Ad.treels -n 0 -zb 10000 -zw -au --undo

}

export -f make_treels
make_treels


# All: unconstrained ${SUPERMATRIX} C60

iqtree2 -s $REF_ALN -m LG+C60+F+G -fs $REF_SITEFREQ --seqtype AA --prefix
SITEHET.${SUPERMATRIX}.constrained_to_topo_d -te
./scratch/tree_building/${SUPERMATRIX}/SITEHET.${SUPERMATRIX}.treefile -ntmax $CPUS -nt $CPUS
-z SITEHET.${SUPERMATRIX}.constrained_to_topo_d.treels -n 0 -zb 10000 -zw -au
```

7.4 Likelihood mapping

```
CPUS=$1

SUPERMATRIX=SOGall

TMPDIR=./scratch/tree_building/${SUPERMATRIX}
mkdir -p $TMPDIR
cd $TMPDIR


var1=$2

var2=$(echo "$var1" | awk '{print tolower($0)}')
var=${var1}_quartets

echo $var2

iqtree2 -S ${SUPERMATRIX}_data -p ${SUPERMATRIX}.partition.nex --seqtype AA --prefix
EDGEPROP.quartet_map_${var2}.${SUPERMATRIX} -lmclust ./data/trees/quartet_map_${var2}.nex
-ntmax $CPUS -nt $CPUS -n 0 -redo
```